

Using Machine Learning to Classify Skin Lesions

DS 4002 Case Study by Katie Lai



Skin health plays an important role in overall well-being, yet many people delay seeking medical attention due to long wait times, high costs, or limited access to specialists. A simple visit to a dermatologist can take weeks to schedule and may involve expensive lab work just to receive a diagnosis. Because of these barriers, early detection of serious conditions, such as skin cancer, is often delayed.

At the same time, advances in machine learning have made it possible to analyze medical images and identify patterns that may not be immediately visible to the human eye. These technologies have the potential to support healthcare professionals and provide more accessible tools for early screening. While these models are not a replacement for medical expertise, they can serve as a helpful first step in identifying whether further evaluation is needed.

In this case study, you will take on the role of a data scientist tasked with developing machine learning models to classify common pigmented skin lesions using dermoscopic images. You will explore how models can learn from image data and investigate whether incorporating additional information, such as patient characteristics, can improve predictions.

Your goal is to build and evaluate models that distinguish between different types of skin lesions and to consider how your modeling choices impact performance. As you work through this case study, you will analyze the data, develop models, and interpret your results in the context of a real-world healthcare application.

In doing this, you will create a GitHub repository that communicates your approach, results, and key insights. Your final product should clearly demonstrate how machine learning can be applied to classify skin lesions and support decision-making in a healthcare setting.

GitHub Link: <https://github.com/katherinejlai/skin-lesions-case-study>